Gaussian Differential Privacy Curve

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Definition

For $\mu \geq 0$, the Gaussian differential privacy tradeoff curve is

$$\beta_\mu(\alpha) = \Phi\left(\Phi^{-1}(1-\alpha) - \mu\right).$$

Its privacy profile is

$$\delta_\mu(\epsilon) = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - e^\epsilon \Phi\left(-\frac{\epsilon}{\mu} - \frac{\mu}{2}\right),$$

with the value at $\mu = 0$ defined by continuity as zero.

Composition

Independent Gaussian privacy losses compose by addition of their squared parameters. Thus, for curves with parameters μ_i,

$$\mu_{\text{comp}} = \sqrt{\sum_i \mu_i^2}.$$

The implementation encloses each arithmetic operation with directed interval arithmetic and rejects a non-finite resulting parameter.